

ALİŞTIRMALAR 4 - TÜREV

1. $f(x) = \arccos \frac{x-5}{2} + \log(6-x) + \sin \sqrt[3]{x-2}$ fonksiyonunun tanım kümesini bulunuz.

$$\bullet \arccos \frac{x-5}{2} \begin{cases} -1 \leq \frac{x-5}{2} \leq 1 \\ -2 \leq x-5 \leq 2 \\ 3 \leq x \leq 7 \end{cases}$$

$$-1 \leq \frac{x-5}{2} \leq 1$$

$$6-x > 0$$

$$6 > x$$

$$-2 \leq x-5 \leq 2$$

$$[3, 6)$$

$$\boxed{3 \leq x \leq 7}$$

$$\bullet \log(6-x) \begin{cases} 6-x > 0 \\ 6 > x \end{cases}$$

$$\bullet \sin \sqrt[3]{x-2} \begin{cases} \mathbb{R} \end{cases}$$

$$D(f) = [3, 6)$$

2. $f(x) = \arcsin(1-x) + \ln(\ln x)$ fonksiyonunun tanım kümesini bulunuz

$$\bullet \arcsin(1-x) \begin{cases} -1 \leq 1-x \leq 1 \\ -2 \leq -x \leq 0 \\ 0 \leq x \leq 2 \end{cases}$$

$$-1 \leq 1-x \leq 1$$

$$\ln x > 0$$

$$-2 \leq -x \leq 0$$

$$x > 0$$

$$0 \leq x \leq 2$$

$$x > 1$$

$$\bullet \ln(\ln x) \begin{cases} \ln x > 0, & x > 0 \\ x > 1 \end{cases}$$

$$(1, 2]$$

$$D(f) = (1, 2]$$

$$3. \lim_{x \rightarrow e} \frac{\ln x - 1}{x - e} = ?$$

$$\frac{\frac{1}{x}}{1} = \frac{1}{x} = \frac{1}{e}$$

$$\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e} = \lim_{x \rightarrow e} \frac{\ln x - \ln e}{x - e} = \lim_{x \rightarrow e} \frac{\ln\left(\frac{x}{e}\right)}{x - e}$$

$$= \lim_{x \rightarrow e} \frac{\ln\left(\frac{x}{e}\right)}{e\left(\frac{x}{e} - 1\right)}$$

$$\frac{x}{e} - 1 = t \Rightarrow x \rightarrow e \Rightarrow t \rightarrow 0$$

$$= \lim_{t \rightarrow 0} \frac{1}{e} \cdot \frac{\ln(t+1)}{t} = \frac{1}{e} \lim_{t \rightarrow 0} \ln(t+1)^{1/t}$$

$$= \frac{1}{e} \ln\left(\underbrace{\lim_{t \rightarrow 0} (t+1)^{1/t}}_e\right) = \frac{1}{e} \cdot \ln e = \frac{1}{e}$$

$$\left[\begin{array}{l} \lim_{n \rightarrow 0} (1+n)^{1/n} = e \\ \lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k \\ \lim_{n \rightarrow 0} \frac{\ln(1+n)}{n} = 1 \end{array} \right]$$

$$4. \lim_{x \rightarrow \frac{\pi}{4}} \frac{\ln(\tan x)}{1 - \cot x} = ?$$

$$\tan x = u \\ x \rightarrow \frac{\pi}{4} \Rightarrow u \rightarrow 1$$

$$\lim_{u \rightarrow 1} \frac{\ln u}{1 - \frac{1}{u}} = \lim_{u \rightarrow 1} u \cdot \frac{\ln u}{u - 1}$$

$$u - 1 = t \\ u \rightarrow 1 \Rightarrow t \rightarrow 0$$

$$= \lim_{t \rightarrow 0} (t+1) \cdot \frac{\ln(t+1)}{t}$$

$$= \underbrace{\lim_{t \rightarrow 0} (t+1)}_1 \cdot \underbrace{\lim_{t \rightarrow 0} \frac{\ln(t+1)}{t}}_1$$

$$= \frac{1}{2}$$

$$\frac{(\tan x)'}{(1 - \cot x)'}$$

$$\frac{\frac{1}{1+\tan^2 x} \cdot (\tan x)'}{\frac{1}{1+\tan^2 x}} \quad \frac{\pi}{4} \quad \cot x$$

$$\cot \frac{\pi}{4} = \frac{1}{1}$$

$$\frac{1}{2}$$

$$5. \lim_{x \rightarrow -1} \frac{\sqrt{\pi} - (\arccos x)'}{(\sqrt{x+1})'} = ?$$

$$\arccos x = t \quad x \rightarrow -1 \Rightarrow t \rightarrow \pi$$

$$x = \cos t$$

$$\frac{\frac{+1}{\sqrt{1-x^2}}}{2 \arccos x} = \frac{1}{2 \sqrt{x+1}}$$

$$\frac{x \cdot \sqrt{x+1}}{x \sqrt{1-x^2} \cdot \sqrt{\arccos x}}$$

$$\sqrt{\frac{1}{2\pi}} \quad \sqrt{\frac{+1}{t \cdot 2 \cdot \pi}}$$

$$\sqrt{\frac{x+1}{1-x^2} \cdot \arccos x}$$

$$\sqrt{\frac{x+1}{-(x+1)(x-1)} \arccos x}$$

$$\sqrt{\frac{-1}{(x-1)(\arccos x)}}$$

$$= \lim_{t \rightarrow \pi} \frac{\sqrt{\pi} - \sqrt{t}}{\sqrt{\cos t + 1}} \cdot \frac{(\sqrt{\pi} + \sqrt{t})}{(\sqrt{\pi} + \sqrt{t})} = \lim_{t \rightarrow \pi} \frac{\pi - t}{\sqrt{2 \cos^2 \frac{t}{2}}} \cdot \frac{1}{(\sqrt{\pi} + \sqrt{t})}$$

$$\pi - t = u$$

$$t \rightarrow \pi \Rightarrow u \rightarrow 0$$

$$= \lim_{u \rightarrow 0} \frac{u}{\sqrt{2} \cdot \cos(\frac{\pi}{2} - \frac{u}{2})} \cdot \frac{1}{(\sqrt{\pi} + \sqrt{\pi + u})}$$

$$= \lim_{u \rightarrow 0} \frac{\frac{u}{2} \cdot 2}{\sqrt{2} \cdot \sin \frac{u}{2}} \cdot \frac{1}{(\sqrt{\pi} + \sqrt{\pi + u})}$$

$$= \frac{1}{\sqrt{2\pi}}$$

$$6. f(x) = \begin{cases} \frac{\ln(\sin x)}{\cos^2 x}, & x \neq \frac{\pi}{2} \\ -\frac{1}{2}, & x = \frac{\pi}{2} \end{cases} \quad x = \frac{\pi}{2} \text{ de sürekli midir?}$$

$$\frac{\cos x}{\sin x} = \frac{1}{-2 \cos x \sin x}$$

$$\frac{\cos x}{\sin x} \cdot \frac{1}{-2 \cos x \sin x} = \frac{-1}{2 \sin^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{\cos^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(1 - \cos^2 x)^{1/2}}{\cos^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{2} \frac{\ln(1 - \cos^2 x)}{\cos^2 x}$$

$$\cos^2 x = -t$$

$$x \rightarrow \frac{\pi}{2} \Rightarrow t \rightarrow 0$$

$$= \lim_{t \rightarrow 0} -\frac{1}{2} \frac{\ln(1+t)}{t}$$

$$= -\frac{1}{2} \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = -\frac{1}{2} = f\left(\frac{\pi}{2}\right) \text{ olduğundan}$$

$f(x)$ sürekli dir.

7. $y = \ln(\sin x)$, $x = \sqrt{\arccos 2^{-3t}}$ ise $\frac{dy}{dt} = ?$ $\frac{1}{2^{3t}}$

☆ $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$ $\frac{\cos x}{\sin x}$ $\frac{dy}{dx} \cdot \frac{dx}{dt}$ 2^{-3x} 2^{-3t}

$$= \frac{\cos x}{\sin x} \cdot \frac{3 \cdot \ln 2 \cdot 2^{-3t}}{\sqrt{1-2^{-6t}}} \cdot \frac{1}{2 \sqrt{\arccos 2^{-3t}}} \cdot \frac{-3 \cdot 2^{-3t} \cdot \ln 2}{\sqrt{1-2^{-6t}}} \cdot \frac{1}{2 \sqrt{\arccos 2^{-3t}}}$$

8. $f(x) = e^{\pi x} - \pi x$ eğrisine teğet olan yatay doğruları bulunuz.

$$f'(x) = e^{\pi x} \cdot \pi - \pi$$

$$\pi(e^{\pi x} - 1) = 0$$

$$f'(x) = \pi \cdot e^{\pi x} - \pi$$

$$(0, 1)$$

yatay doğruların eğimi "0 (sıfır)" olduğundan,

$$e^{\pi x} = 1$$

$$x = 0$$

$$\pi(e^{\pi x} - 1) = 0 \Rightarrow e^{\pi x} = 1 \Rightarrow \pi x = 0 \Rightarrow \underline{x = 0}$$

$$x = 0 \Rightarrow f(x) = y \Rightarrow y = e^0 = 1 \Rightarrow \underline{y = 1} \text{ doğrusudur.}$$

9. $\cos(3x+y) + \sin(x+3y) = -1$ eğrisinin $A(0, \frac{\pi}{2})$ noktasındaki

$$-\sin(3x+y) \cdot (3+y') + \cos(x+3y) \cdot (1+3y') = 0$$

teğet ve normal doğru denklemlerini bulunuz.

$$-\sin \frac{\pi}{2} \cdot (3+y') + \cos \frac{3\pi}{2} \cdot (1+3y') = 0$$

$$-(3+y') \cdot \sin(3x+y) + (1+3y') \cdot \cos(x+3y) = 0$$

$$-3-y' = 0 \quad y' = -3$$

$$A(0, \frac{\pi}{2}) \Rightarrow -(3+y') \cdot \sin \frac{\pi}{2} + (1+3y') \cdot \cos(3\frac{\pi}{2}) = 0$$

$$-(3+y') = 0 \Rightarrow y' = -3 = m_T, m_N = \frac{1}{3}$$

$$y = -3x + \frac{\pi}{2}$$

$$y = \frac{x}{3} + \frac{\pi}{2}$$

T.D.D. : $y - \frac{\pi}{2} = -3(x-0) \Rightarrow \underline{y = -3x + \frac{\pi}{2}}$ //

N.D.D. : $y - \frac{\pi}{2} = \frac{1}{3}(x-0) \Rightarrow \underline{y = \frac{1}{3}x + \frac{\pi}{2}}$ //

10. $(1,002)^3 - 2\sqrt{1,002} + 3$ yaklaşık değerini lineerleştirme ve diferansiyel yardımıyla bulunuz.

$$f(x) = x^3 - 2\sqrt{x} + 3 \quad a=1, \quad f(1)=2$$

$$f'(x) = 3x^2 - \frac{1}{\sqrt{x}} \quad , \quad f'(1)=2$$

$$a) f(x) \approx L(x) = f(1) + f'(1)(x-1) = 2 + 2 \cdot (x-1)$$

$$f(1,002) \approx L(1,002) = 2 + 2 \cdot 0,002 = 2,004$$

$$b) dy \approx \Delta y \quad \Delta x = 0,002 = dx$$

$$dy = f'(x) \cdot dx = f'(x) \cdot \Delta x = f'(1) \cdot 0,002 = 0,004$$

$$\Delta y = f(1,002) - f(1) = f(1,002) - 2$$

$$dy \approx \Delta y \Rightarrow f(1,002) \approx 2 + 0,004 = 2,004$$

11. $f(x) = \left(\tan \frac{x}{2}\right)^{x \arcsin 2x}$ ise $f'(x) = ?$

$$\ln f(x) = (x \cdot \arcsin 2x) \cdot \ln \left(\tan \frac{x}{2}\right)$$

$$\frac{f'(x)}{f(x)} = \left(\arcsin 2x + \frac{2x}{\sqrt{1-4x^2}}\right) \cdot \ln \left(\tan \frac{x}{2}\right) + x \cdot \arcsin 2x \cdot \frac{\frac{1}{2} \cdot \sec^2 \frac{x}{2}}{\tan \frac{x}{2}}$$

$$f'(x) = \left(\tan \frac{x}{2}\right)^{x \arcsin 2x} \cdot \left[\left(\arcsin 2x + \frac{2x}{\sqrt{1-4x^2}}\right) \cdot \ln \left(\tan \frac{x}{2}\right) + x \cdot \arcsin 2x \cdot \frac{\sec^2 \frac{x}{2}}{2 \tan \frac{x}{2}} \right]$$

$$\ln f(x) = x \cdot \arcsin 2x \cdot \left(\ln \tan \frac{x}{2}\right)$$

$$\frac{f'(x)}{f(x)} =$$

12. g ve h fonksiyonları $g(1)=h'(1)=1$, $g'(1)=h(1)=2$ şartlarını sağlayan pozitif değerli ve türevlenebilen birer fonksiyon olmak üzere f fonksiyonu da $f(x)=[g(x^2)]^{h(x)}$ ile tanımlı olsun. $f'(1)$ değerini bulunuz.

$$f(x)=[g(x^2)]^{h(x)}, f(1)=(g(1))^{h(1)}=1^2=1$$

$$\ln f(x) = h(x) \cdot \ln(g(x^2))$$

$$\frac{f'(x)}{f(x)} = h'(x) \cdot \ln(g(x^2)) + h(x) \cdot \frac{2x \cdot g'(x^2)}{g(x^2)}$$

$$f'(1) = \underbrace{f(1)}_1 \cdot \left[\underbrace{h'(1)}_1 \cdot \underbrace{\ln(g(1))}_0 + \underbrace{h(1)}_2 \cdot \frac{2 \cdot 1 \cdot g'(1)}{\underbrace{g(1)}_1} \right] = 2 \cdot 2 \cdot 2 = 8$$

13. $f(x) = 2^{x^2+\cos x} + 3^{x \ln(x+1)}$ ise $f'(x) = ?$

$$f'(x) = (2x - \sin x) \cdot \ln 2 \cdot 2^{x^2+\cos x} + \ln 3 \cdot 3^{x \ln(x+1)} \cdot \left(\ln(x+1) + \frac{x}{x+1} \right)$$

$$2^{x^2+\cos x} \cdot (2x - \sin x) \cdot \ln 2 + 3^{x \ln(x+1)} \cdot \ln 3 \cdot \left(\ln(x+1) + \frac{x}{x+1} \right)$$

14. $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x\sqrt{8+x^2}$ fonksiyonunun tersinin mevcut olduğunu gösterip $(f^{-1})'(3)$ değerini bulunuz. $\frac{16x+4x^3}{2\sqrt{8x^2+x^4}} \quad \frac{20}{6} =$

$$f'(x) = \sqrt{8+x^2} + x \cdot \frac{2x}{2\sqrt{8+x^2}} = \frac{2x^2+8}{\sqrt{8+x^2}} > 0 \quad \forall x \in \mathbb{R} \text{ için } f'(x) > 0$$

olduğundan f artan

$$x\sqrt{8+x^2} = 3 \Rightarrow x=1$$

$$\frac{1}{f'(f^{-1}(3))}$$

$$\frac{1}{f'(1)}$$

$$\Rightarrow f, 1-1$$

$$\frac{6}{20} \Rightarrow f \text{ 'in tersi mevcuttur.}$$

$$(f^{-1})'(3) = \frac{1}{f'(1)} = \frac{1}{\frac{10}{3}} = \frac{3}{10}$$

$$\left(\frac{3}{10} \right)$$

(Fonksiyonun tanım kümesi üzerinde çalışıldığı için örtenlik sağlanır.)

15. $y = (s+3)^2$, $s = \sqrt{t-3}$, $t = x^2$ ise $\frac{dy}{dx} \Big|_{x=2} = ?$

$$\frac{dy}{dx} = \frac{dy}{ds} \cdot \frac{ds}{dt} \cdot \frac{dt}{dx}$$

$$= 2 \cdot (s+3) \cdot \frac{1}{2\sqrt{t-3}} \cdot 2x$$

$$\frac{dy}{dx} \Big|_{x=2} = 2 \cdot (1+3) \cdot \frac{1}{2 \cdot 1} \cdot 2 \cdot 2 = 16$$

$$x=2 \Rightarrow t=4 \Rightarrow s=1$$

$$2(s+3) \cdot \frac{1}{2\sqrt{t-3}} \cdot 2x$$

$$2 \cdot 4 \cdot \frac{1}{2} \cdot 4 = 16$$

16. $\tan(xy) = xy \Rightarrow y' = ?$

$$(y + xy') \cdot \sec^2(xy) = y + xy'$$

$$xy' \cdot \sec^2(xy) - xy' = y - y \cdot \sec^2(xy)$$

$$y' (x \cdot \sec^2(xy) - x) = y (1 - \sec^2(xy))$$

$$y' = \frac{y \cdot (1 - \sec^2(xy))}{-x \cdot (1 - \sec^2(xy))}$$

$$y' = -\frac{y}{x}$$

$$\sec^2(xy) \cdot (y + xy') = y + xy'$$

$$xy' \cdot \sec^2(xy) + y \sec^2(xy) = y + xy'$$

$$xy' (\sec^2(xy) - 1) = -y \sec^2(xy)$$

$$(\sec^2(xy) \neq 1)$$

$$y' = \frac{-y \cdot \sec^2(xy)}{x \cdot (\sec^2(xy) - 1)}$$

17. $f(x) = \frac{1-x}{1+x}$ ise $f^{(n)} = ?$

$$f'(x) = \frac{-(1+x) - (1-x)}{(1+x)^2} = \frac{-1-x-1+x}{(1+x)^2} = -2 \cdot (1+x)^{-2}$$

$$f''(x) = (-2) \cdot (-2) \cdot (1+x)^{-3}$$

$$f'''(x) = (-2) \cdot (-2) \cdot (-3) \cdot (1+x)^{-4}$$

$$f^{(4)}(x) = (-2) \cdot (-2) \cdot (-3) \cdot (-4) \cdot (1+x)^{-5}$$

⋮

$$f^{(n)}(x) = 2 \cdot n! \cdot (-1)^n \cdot (1+x)^{-(n+1)}$$

$$f(x) = \frac{1-x}{1+x}$$

$$f'(x) = \frac{-2}{(1+x)^2}$$

$$f''(x) = \frac{-2 \cdot 2}{(1+x)^3}$$

$$-1(1+x) - (1-x)$$

$$-1-x-1+x$$

$$-2(1+x)^{-2}$$

$$\frac{-2 \cdot 2}{(1+x)^3}$$

$$f^{(3)}(x) = \frac{-2 \cdot 2 \cdot 3}{(1+x)^4}$$

$$\frac{2 \cdot n! \cdot (-1)^n}{(1+x)^{n+1}}$$

18. $f(x) = [\cos(x^4)]^{\arctan x^2}$ fonksiyonu türemlenebilen bir fonksiyon olmak üzere $f'(0) = ?$ 0

$$\ln f(x) = \arctan x^2 \cdot \ln \cos x^4$$

$$\frac{f'(x)}{f(x)} = \frac{2x}{1+x^4} \cdot \ln \cos x^4 + \arctan x^2 \cdot \frac{-\sin x^4 \cdot 4x^3}{\cos x^4}$$

$$\ln f(x) = \arctan x^2 \cdot \ln (\cos(x^4))$$

$$\frac{f'(x)}{f(x)} = \frac{2x}{1+x^4} \cdot \ln(\cos(x^4)) - \arctan x^2 \cdot \frac{4x^3 \cdot \sin(x^4)}{\cos(x^4)}$$

$$f'(0) = f(0) \cdot [0 - 0] = 0$$

19. $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ fonksiyonunun artan ve azalan olduğu aralıkları belirleyiniz.

$$f'(x) = 12x^3 - 12x^2 - 24x = 0$$

$$x^3 - x^2 - 2x = 0$$

$$x(x^2 - x - 2) = 0$$

$$x(x-2)(x+1) = 0$$

x	$-\infty$	-1	0	2	∞
$f'(x)$	-	+	-	+	
$f(x)$	$\searrow$	$\nearrow$	$\searrow$	$\nearrow$	

artan : $(-1, 0) \cup (2, \infty)$

azalan : $(-\infty, -1) \cup (0, 2)$

20. $y = x^3 - 2x^2 + x - 2$ eğrisinin hangi noktadaki teğeti x-eksenine paraleldir.

$$y' = 3x^2 - 4x + 1, \text{ teğet x-eksenine paralel ise } \frac{dy}{dx} = 0 \text{ olmalıdır.}$$

$$3x^2 - 4x + 1 = 0$$

$$(x-1)(3x-1) = 0$$

$$\Rightarrow x=1 \Rightarrow y=-2$$

$$\Rightarrow 3x-1=0 \Rightarrow x=\frac{1}{3} \Rightarrow y=-\frac{50}{27}$$

$$A(1, -2), B(\frac{1}{3}, -\frac{50}{27})$$